



Integrating AI ethics in wildlife conservation AI systems in South Africa: a review, challenges, and future research agenda

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Abstract

With the increased use of Artificial Intelligence (AI) in wildlife conservation, issues around whether AI-based monitoring tools in wildlife conservation comply with standards regarding AI Ethics are on the rise. This review aims to summarise current debates and identify gaps as well as suggest future research by investigating (1) current AI Ethics and AI Ethics issues in wildlife conservation, (2) Initiatives Stakeholders in AI for wildlife conservation should consider integrating AI Ethics in wildlife conservation. We find that the existing literature weakly focuses on AI Ethics and AI Ethics in wildlife conservation while at the same time ignores AI Ethics integration in AI systems for wildlife conservation. This paper formulates an ethically aligned AI system framework and discusses pre-eminent on-demand AI systems in wildlife conservation. The proposed framework uses agile software life cycle methodology to implement guidelines towards the ethical upgrade of any existing AI system or the development of any new ethically aligned AI system. The guidelines enforce, among others, the minimisation of intentional harm and bias, diversity in data collection, design compliance, auditing of all activities in the framework and ease of code inspection. This framework will inform AI developers, users, conservationists, and policymakers on what to consider when integrating AI Ethics into AI-based systems for wildlife conservation.

Keywords Wildlife conservation concerns · Human–wildlife conflicts · AI Ethics · AI in wildlife conservation · AI Ethics integration · Artificial intelligence

1 Introduction

The topic of Artificial Intelligence (AI) Ethics is an emerging and growing field. AI Ethics focuses on ethical principles, guidelines, laws, and policies that are AI-related. Ethical AI guidelines focus on the ethical use of data, algorithms, tools, and technologies. Some of the currently implemented AI Ethics in wildlife conservation discussed in Sect. 1.1 may apply to other sectors of AI applications.

1.1 Current AI ethics in wildlife conservation

The security, quality, and transparency of data and algorithms as well as data use and information disclosure are the core goals in most AI Ethics guidelines. The emphasis is usually on minimising bias in data and promoting inclusiveness by collecting diverse and equally represented data sets (Ienca and Vayena 2019). Reducing bias in data sets is a challenge when designing equitable algorithms (Parkes and Vohra 2019). Minimising these biases is possible using quality data sets attained through copyrighting of original data sets and improved spatial resolution (Jantke et al. 2013). With increased standards, these quality data sets become high in demand among data providers and AI developers. Wildlife conservation data providers such as the Global Biodiversity Information Facility (GBIF) have enforced data quality and protection through copyrighting and licensing the data sets belonging to them (GBIF 2020). In South Africa, data providers such as the Endangered Wildlife Trust (EWT) emphasise respect for embargoes and sensitivities of data sets (EWT 2020). These requirements from providers of

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data sets lead to fair use of data (Kaeser-Chen et al. 2019). Fairness in conservation data and algorithms is possible when stakeholders observe some of the standards below:

- Agreeing to the terms of use before data provision. The protected and conservation areas database by the Department of Environmental Affairs (DEA) in South Africa observes this standard. Users are required to agree to the terms of data usage before data access and provisions are granted (DEA 2020).
- Data providers should document guidelines to users on the ethical use of the data sets; the GBIF implements this standard (GBIF 2020). In these documented guidelines, conservationists should show how large data sets are processed and how AI models operate on the data. The significant growth in digital conservation data sets further motivates the latter (Welz 2017).
- Justification of intentions before data access is provided. Stating reasons for using the requested data is vital, for example, a person or entity requesting conservation data must justify to the conservation agency what they intend to use the data for. These applicants must convincingly show the interplay between AI platforms, animals, and conservationists in their submissions.
- Documentation of known biases and release notes from published data sets and models.
- Explanation of machine learning algorithms (Doshi-Velez et al. 2017).
- Enforcing transparency on data access leads to increased trust in the AI models used Felzmann et al. (2019). Microsoft AI for Earth enforces transparency by developing and deploying machine learning models which do not cause harm to the human race and its environment. It emphasises fundamentals such as imposing AI systems to remain friendly to the human race as well as incorporating the natural environment's health (World Economic Forum 2018).

1.2 Research gaps on the practical implementation of Ethical AI in wildlife conservation

Globally, the death rate of large animals is highly attributed to anthropogenic activities (Gantchoff et al. 2020). The endangered large animals considered in this review are the African elephant, rhino, zebra, cheetah, African wild dog, leopard, lion, antelope, and giraffe on the Red List (animals in South Africa). In Africa, cheetah cubs have an infant mortality rate of 95% (Science News Staff 2020). In 2020, the African wild dog population was close to 600,000 across Africa. Their population has now declined to between 3000 and 5000 animals (Loparo 2020). The African Wildlife Foundation (AWF) states that globally, the leopard population has declined by 31% over the last 22 years. The

AWF argues further that the African lion population has also declined by a third over the past 20 years. In addition, the zebra and giraffe population has declined by 54% and 40%, respectively, over the last 30 years (AWF 2020a, b). In Kenya, antelopes dropped in numbers from 202 in 1976 to 19 in 2019 (Kimanzi 2020). Rhino poaching grew by 9,200% between 2007 and 2014. In 2014 alone, there were 1,215 rhinos killed in South Africa (AWF Emergency Response 2020). The poaching of African elephants is predominantly for ivory, and close to 40,000 are killed every year (Wittemyer et al. 2014). AI-based monitoring systems have been adopted to mitigate the alarming death rate of large animals.

AI systems perform image processing, speech recognition, decision-making, and natural language processing (Lecun et al. 2015). AI-based systems require human-like intelligence and have solved both toy and real-world problems (Barr and Feigenbaum 2014). These AI systems in wildlife conservation are increasingly applied:

- To reduce the high death rates of large animals;
- To improve biodiversity monitoring and conservation (Vinuesa et al. 2020);
- For prevention of human–wildlife conflicts;
- For saving the lives of large animals in wild reserves, through fast and accurate wildlife monitoring and well-organised data analysis (Cookson 2019); and
- To mitigate biodiversity loss and alterations in climate change.

Integration of AI Ethics in AI for wildlife conservation has so far received less attention, as most of the existing guidelines only focus on the conservation or AI Ethics domains. In Sect. 2, we discuss AI Ethics concerns in wildlife conservation. We conducted a non-exhaustive review of existing AI monitoring tools, laws, and policies in South Africa. A sample of these wildlife conservation concerns are:

- Poaching incidences, where large animals are hunted down for consumption, illegal trade and medicine (AWF 2020a, b);
- Cyber poaching, where poachers hack into global positioning system (GPS) tags and steal geospatial data to track and kill animals;
- Violation of animal rights, and the relative unfairness embedded in the South African Meat Safety Act of 2000 (the Act regulates the slaughtering of a range of large animals);
- Legal barriers and ethical constraints are associated with security and privacy issues raised by the use of drones. In South Africa, the use of drones inside national parks is prohibited when the operator is unlicensed and unauthorised (SANParks 2015);

- Lack of enforcement resources by rangers as they conduct patrols in uninhabited protected areas. When resources are not available to track poacher routes, rangers or conservationists become vulnerable to poachers and predators;
- Illegal trade on social media and its associated insecurities (Kerry 2012); and
- Collar tag side effects, such as alternations in animal behaviour due to the transmitter effects.

The AI Ethics concerns above are also prevalent in South Africa. This prevalence is partially due to (1) the non-existence of guidelines that focus on AI Ethics, (2) the lack of normative values, and (3) the limited implementation of these guidelines into AI tools and technologies (Hagendorff 2020). To date, wildlife conservation agencies aspire to solve the above ethical concerns.

1.3 Contribution

This paper describes AI Ethics and reviews the existing literature on current AI Ethics, as well as on AI Ethical concerns in wildlife conservation. Contemporary AI Ethics in wildlife conservation focuses on understanding AI Ethics applications in the wildlife conservation domain. The identified AI Ethical concerns consist of lack of algorithm audit, bias in data and algorithms, errors in data samples, violation of animal rights, and cyber poaching. We state these concerns to identify the gaps, suggest future research agendas and recommendations for the implementation of AI Ethics in AI for wildlife conservation. The recommended framework, together with guidelines suggest a way to implement and integrate AI Ethics into AI systems for wildlife conservation. In addition, we propose new on-demand AI systems in wildlife conservation.

1.4 Manuscript organisation

The structure of this paper is as follows. Section 2 presents several AI Ethics concerns exacerbated in wildlife conservation. This section aims at identifying ethical issues that arise from AI-based monitoring tools, laws, and policies for AI for wildlife conservation within South Africa. Section 3 discusses the integration of AI Ethics into AI for wildlife conservation and formulates many future research agendas as well as recommendations. We conclude in Sect. 4.

2 The AI ethics concerns exacerbated in wildlife conservation

AI Ethics in wildlife conservation is a domain, where developers and users of AI are concerned with the ethical use of AI platforms for wildlife conservation. The tools and

algorithms for AI systems within wildlife conservation are developed and deployed through close collaborations between conservation policymakers and software companies. AI tools in wildlife conservation monitor, track and protect animals and address some ethical concerns highlighted within the conservation sector (Kaesler-Chen et al. 2019; Murray 2014). The monitoring also mitigates human–wildlife conflicts over land and food (Ihwagi et al. 2015).

This study adopts a methodology that analyses and categorises AI Ethics concerns, conservation laws and policies in wildlife conservation in South Africa. Our review investigates: (1) what the current AI Ethics and AI Ethics issues in wildlife conservation are, and (2) what stakeholders of AI for wildlife conservation should consider when integrating AI Ethics into wildlife conservation. Papers were searched using the following keywords; “AI Ethics,” “Ethics,” “wildlife trafficking” “wildlife monitoring tools,” “conservation laws + South Africa,” “conservation policies + South Africa,” “agile methodology,” “camera traps + South Africa” and “artificial intelligence.” This was the most appropriate methodology given our goal to assist stakeholders in deciding on best practices for integrating AI Ethics into AI for wildlife conservation. This methodology allowed us to use a review question and extract literature on a specific topic to help us identify the gaps.

The reviewed publications were extracted from Google Scholar and Scopus databases. We also extracted information from wildlife conservation, government gazettes or AI-related company reports. We trusted reliable internet sources, such as conservation websites, education websites, and news websites with clear, objective, accurate and relevant information. As the field is not exhaustively explored in academic publications, these reliable internet sources did fill some gaps in the literature. The number of references cited and the sources of reference in this review are visualised in Fig. 1 and their corresponding year(s) of publications are shown in Fig. 2.

2.1 Wildlife monitoring tools and AI ethics concerns in South Africa

Most of the wildlife monitoring tools below are being increasingly used all over Africa (including South Africa). Monitoring of animals in national parks uses tools, such as drones, remote sensing collars and camera traps, among others (Branko et al. 2018). In South Africa, these tools help conservationists to know if the animals are ill or injured. They also keep track of animal movement patterns, variations in demographics, habitat utilisation, breakouts, outbreaks of diseases, and notifications of poaching incidents of the few endangered species on the Red List (Wildlife Act 2020). Although these monitoring tools have recorded

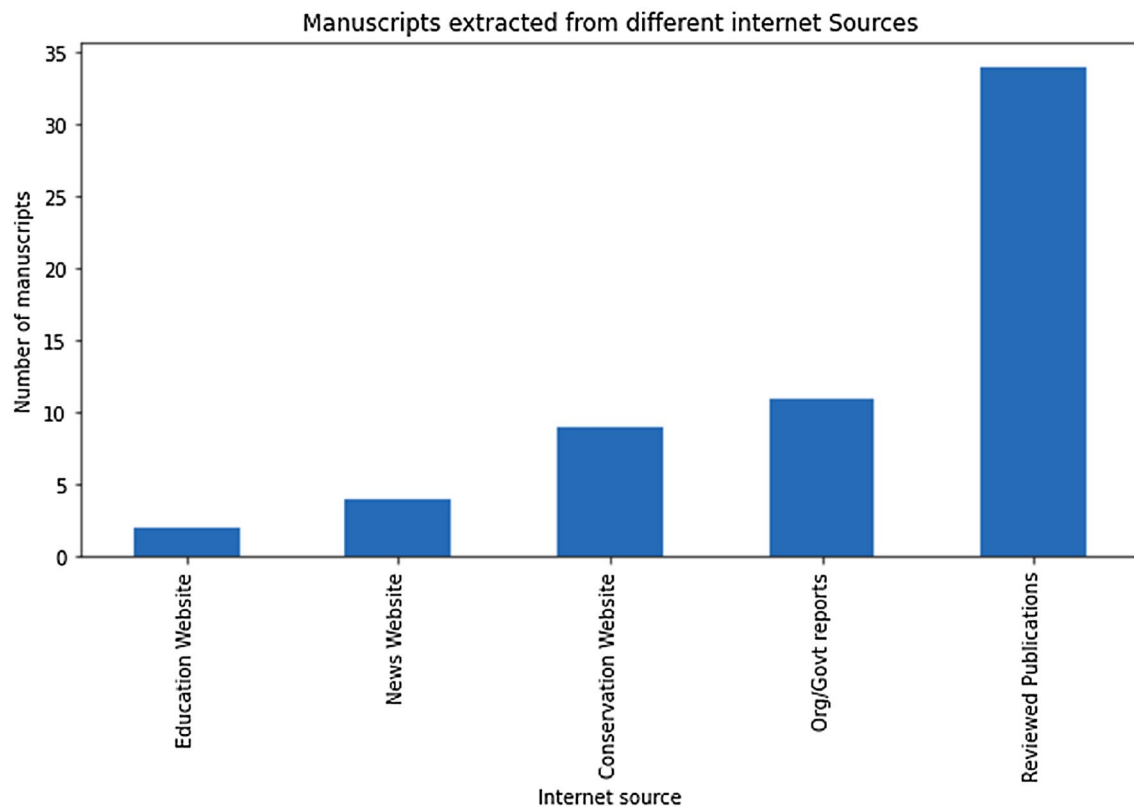


Fig. 1 Number of manuscripts used in this review from each database, for example, reviewed publications (Google Scholar, Scopus) and other internet sources

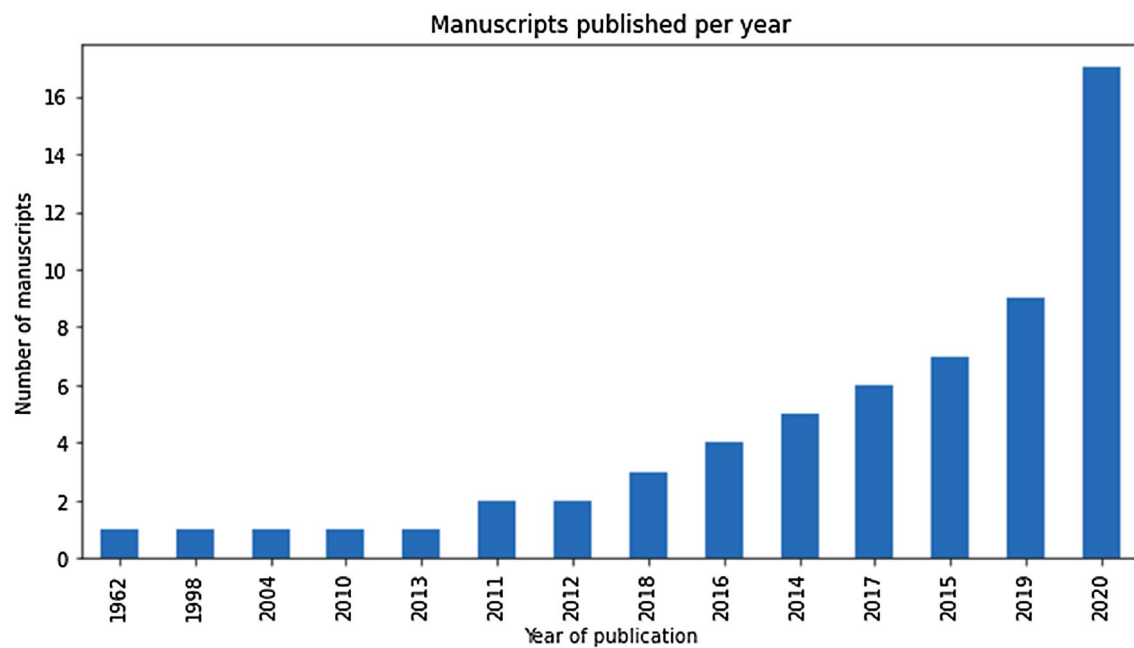


Fig. 2 Year of publications in this review from reviewed publications (Google Scholar, Scopus), and other internet sources

significant results, a research gap still exists about whether these tools incorporate AI Ethics principles. In this section, we identify the dominant AI Ethics concerns in some implemented AI-based wildlife monitoring tools. Lokare et al. (2014) monitored endangered species using footprints to detect animals in wild reserves. They identified and modelled the curve of interest in images; the modelling was based on a stochastic process on a manifold. This manifold approximation shape uses eigen decomposition of previously observed patterns. The model captures the imaging and geometric information of a set of feasible curves. An interest curve selected among these feasible curves finds the path with the highest likelihood generated by the defined stochastic process. The model was tested on the images of tiger footprints with ground-truth segmentation provided by experts in the field. Results showed that the overall average distance error (ADE) for the toes, footprints, and pad was within 8 pixels while considering the outliers. The model computed the ADE between the ground truth points and the results. We identify ethical concerns as bias in data and algorithms in this work. The ethical concerns could be due to the smallness of the data sample, characterised by the unequal representation of features, such as toes and pads. The latter could have affected the outcome of the algorithm represented by the ADE. Auditing each feature and parameter in the algorithms can help in optimising the model.

In African countries, such as Côte d'Ivoire and Tanzania, Kalan et al. (2016) performed the monitoring of chimpanzees using sounds. They used passive acoustic monitoring systems (PAM), which capture sounds at a long-distance produced by wild chimpanzees in their natural habitats. They mounted autonomous recording units on small trees of 1–2 m from the ground. This short distance was selected, because drums and hoots are produced by chimpanzees while travelling on the ground. An algorithm detected and classified chimpanzee sounds by extracting chimpanzee drums from continuous autonomous recording units. Results show that chimpanzee sounds were detected and predicted the direct and indirect indices of chimpanzee presence. Errors in the data set occur often, since PAM can detect long-distance acoustic signals of other large animals, for example, lions and wolves. Such errors are a major AI Ethics concern, as they lead to a misrepresentation of data samples; the existing algorithms cannot uniquely classify and associate the detected sound with the emitting species.

Gonzalez et al. (2016) developed a conservation drone, with a system made of airborne and ground segments. The airborne segment consisted of multirotor unmanned aerial vehicles (UAVs), navigation systems, thermal cameras, gimbal systems, and video transmitters. The ground segment consisted of the ground station software installed on laptops, the data link, and video receivers for remote display and recording. The system captures thermal images

and videos to perform object detection, classification, and wildlife tracking in the forest or open areas. These drones can count and monitor wild animals, and catch poachers in the act. The ethical AI issues associated with this platform are privacy and data security (Cracknell 2017; Stocker et al. 2017). In South Africa, drones are categorised as aircraft. The government uses legislation, such as the Civil Aviation Act, part 101 of the Civil Aviation Regulations, and part 1 of the Civil Aviation Technical Standards to regulate drone activities in the national airspace (Cronjé 2020).

The Zululand Rhino Reserve in northern KwaZulu-Natal (South Africa) used camera traps to estimate the abundance of leopards (Chapman and Balme 2010). The survey used 34 paired camera-trap stations for 40 days. Reliable population estimates for leopards are beneficial to conservation and management as they enable informed decisions for a species that, although widespread, is heavily persecuted. We highlight that there is no transparency in how the images are transmitted, stored, and accessed, which causes a breach of privacy and is prone to poaching incidences. In 2016, integrations of the Spatial Monitoring and Reporting Tool (SMART) and the Protection Assistant for Wildlife Security (PAWS) were successfully implemented. PAWS is a tool that uses AI to predict poaching routes or the location of poaching incidences and helps rangers patrol efficiently. It specifically learns the behaviour models of the poachers from the crime data collected. Using the behaviour model of poachers, PAWS calculates a randomised patrolling strategy in the form of a set of patrol routes and the probabilities of selecting a patrol route. PAWS then suggests patrol routes sampled from this strategy to the patrollers. The first PAWS tests done in Uganda were important and led to the massive deployment of this tool in other parts of Africa (Fang et al. 2016). However, conservation agencies did identify challenges, such as the lack of law enforcement resources to conduct these patrols in uninhabited protected areas. When rangers lack law enforcement resources, for example, gadgets for collecting crime data, system accuracy in predicting poaching routes is poor due to limited data sets. This is an AI Ethics issue.

In 2018, there was an implementation of a web-based SMART tool that monitors, evaluates, and improves the effectiveness of conservation management. The Wildlife Conservation Society (WCS) and World Wildlife Fund (WWF) well as thousands of parks and game reserves use SMART. This tool minimises the risk faced by conservationists as they expose their lives to predators and poachers when protecting biodiversity. Rangers and conservationists use the SMART system to collect, store, communicate, and evaluate range-based data on illegal activities, biodiversity, enforcement routes, management actions, and threat levels. SMART helps rangers to identify hotspots of illegal activities. The tool has increased patrol efforts, reduced poaching,

and increased prey populations (Rawles 2018). This system is a law enforcement monitoring tool, specifically for queries and reports, which are very advanced and systematic. However, the SMART system is centralised and prone to security weaknesses, leading to cyber hacking by poachers. Developers can design or customise the current system to be a distributed system using peer-to-peer networks, such as those based on blockchain technology (Huang et al. 2020). The design can minimise the AI Ethics concern of security. In another work in Tanzania, Resolve company in collaboration with Intel developed the TrailGuard AI system (Dinerstein 2018) which detects a poacher and notifies rangers about their exact location. This system uses intelligent camera vision technology that detects motion. It uses an on-device AI algorithm to analyse images in real time and sends an alert with a photo of identified humans or vehicles in any of the captured frames. An algorithm audit is recommended for this work to identify ethical concerns in algorithms and ensure the system is fair and accountable.

Maroto-Molina et al. (2019) developed GPS and very high frequency (VHF) tags used to monitor wildlife and livestock. These low-cost tags gather, process and transmit the location data of all tagged animals. In South Africa, GPS and VHF tags are also used (Wildlife Act 2014), with GPS showing tremendous results when tested on African elephants (Bindi et al. 2011). The type of animals tracked ranges from medium and small to large animals. Unfortunate, this work ignored ethical AI concerns, such as animal rights violations and collar tag side effects (Gutema 2015). Based on the non-exhaustive survey conducted in Sect. 2.1, we present a sample of significant ethical concerns in Table 1.

2.2 Wildlife conservation laws and policies and ethical concerns in South Africa

In the process of ethically minimising human–wildlife conflicts, stakeholders draft laws and policies. The drafted guidelines help to achieve ethically conscious wildlife conservation tools, algorithms, and data. However, some of these laws and policies contribute to ethical concerns. Governments document policies and rules that govern wildlife conservation practices and they ensure the guidelines favour species. In February 2020, in South Africa, a proposal was tabled to modify the Meat Safety Act 40 of 2000, which introduced a threat to large animals (Center for Environmental Rights 2020). This proposal intended to increase the number of wildlife species that could be slaughtered for human or animal consumption; it made the slaughtering of listed animals legal if done in abattoirs. Other Acts that protect these endangered large animals have had a positive impact on their survival. The Animals Protection Act of 1962 (Republic of South Africa 1962) works towards minimising cruelty to animals. A clause in

this Act stipulates that animal “means any equine, bovine, sheep, goat, pig, fowl, ostrich, dog, cat or other domestic animal or bird, or any wild animal, wild bird or reptile which is in captivity or under the control of any person.” The management and conservation of large animals in South Africa is done through the National Environmental Management: Biodiversity Act of 2004 (Parliament of Republic of South Africa 2004). This Act protects vulnerable and endangered species. In addition, Act No. 62 of 1998: Animal Improvement Act, 1998 provides for the breeding, identification, and utilisation of genetically superior animals to improve the production and performance of animals in the interest of the Republic, as well as to provide for matters connected to that (Republic of South Africa 1998). In Table 2, we summarise these Acts and legislations for some of the conservation laws in South Africa.

Besides laws and policies from the government, conservation policymakers and non-government organisations also contribute to conservation policies. Globally, organisations such as the National Wildlife Federation (NWF) dedicate themselves to follow America's fundamental framework of conservation laws and policies (National Wildlife Federation 2020). The International Union for Conservation of Nature and related institutions have developed policies, regulations, and best practices in the conservation sector regarding connectivity (NWF 2020). Organisations such as the WWF have influenced policies in multiple African countries. The AWF organisation also contributes to effective and sustainable conservation in Africa. It fosters African-led and African-implemented policies by providing African governments and development partners with relevant information to plan, design, and enact policies that support wildlife and wildlands (AWF 2020a, b). In 2015, the South African Government published a notice called the Biodiversity Economy Strategy, covering the period up to 2030. The policy and strategy document seeks to increase the biodiversity contribution to the gross domestic product while preserving the country's ecosystem. It focuses on, for example, enhancing growth in both the wildlife and tourism sectors (Republic of South Africa 2015).

Most laws and policies contribute to the protection of a large range of species. However, a number of them such as the South African Meat Safety Act of 2000 does the opposite. In addition, a majority of South African laws and policies reviewed do not sufficiently provide guidelines on the usage of AI-based monitoring tools in wildlife conservation. The latter contributes to AI Ethics concerns highlighted in this section. Table 2 shows a representative sample of major South African laws and policies in wildlife conservation.

Table 1 Representative sample of wildlife monitoring tools and related AI ethics

Authors	Year	Tool	Data	Results	AI Ethics concerns
Lokare et al. (2014)	2014	Animal identification via footprint recognition	Footprint images	The overall ADE for the toes, pad, and entire footprint is within 8 pixels (considering the outliers) The model captures both imaging and geometric information for the interest curve	Bias in data and algorithms
Kalan et al. (2016)	2016	Passive acoustic monitoring	Acoustic sounds	An algorithm detected and classified chimpanzee sounds by extracting chimpanzee drums from continuous autonomous recording units Predictions of chimpanzee sounds by direct and indirect indices of chimpanzee presence	Data samples are prone to errors, since PAM systems can detect long-distance acoustic signals of other large animals
Gonzalez et al. (2016)	2016	UAVs/Drones	Thermal image and video recordings	Object detection Classification of animal species Tracks wildlife Drones can count and monitor wild animals and catch poachers in the act	Experience legal barriers due to privacy and safety concerns (Cracknell 2017; Stocker et al. 2017)
Fang et al. (2016)	2016	PAWS	Human and animal signs of activity	Predicts where poachers will strike next by understanding where and when poacher attacks have occurred Total patrolling time is limited Patrollers can only move to nearby areas Limited law enforcement resources to conduct patrols	Lack of law enforcement resources to conduct these patrols in uninhabited protected areas makes them vulnerable to poachers and predators
Rawles (2018)	2018	SMART	Data on: illegal activities, biodiversity, enforcement routes, management actions, and threat levels	SMART identifies hotspots of illegal activities Increased patrol effort Reduced poaching of these large animals Increased biodiversity populations It implements law enforcement, since it is a law enforcement monitoring tool; specifically regarding queries and reports, hence data privacy	The high risk faced by conservationists exposing their lives to predators and poachers as they protect biodiversity. The weakness is due to the foreseen risk of security of the current system

Table 1 (continued)

Authors	Year	Tool	Data	Results	AI Ethics concerns
Dinerstein (2018)	2018	TrailGuard AI/camera traps	Images of animals, humans or cars	The system detects a poacher and notifies rangers about the exact location of the poachers Uses smart camera vision technology that detects motion Alerts those in charge when humans or vehicles are detected Some level of data security is observed	Lack of algorithm audit
Maroto-Molina et al. (2019)	2019	Low-cost Internet of Things (IoT)-based System	Location data	Developed a low-cost, collar tagging system to monitor animals Can gather, transmit, and process the location data of all tagged animals A datacentre that limits data access for research and public use	Violation of animal rights through collar tag effects (Gutema 2015)
Frey et al. (2017)	2017	Wildlife monitoring centre	Geospatial data		Cyber—Poaching, which has enabled poachers to hack into GPS tags and steal geospatial data to track and kill animals
Chapman and Balme (2010)	2020	Camera traps	images	Six individual adult leopards (three males and three females) were photographed on 19 occasions during the survey	No transparency on how the images are transmitted, stored and who has access to them. This causes a privacy breach

Table 2 Representative sample of conservation laws that affect large animals in South Africa

References	Year	Act	Legislation
Republic of Africa (Republic of South Africa 1962)	1962	Animals Protection Act, 1962	To consolidate and amend the laws relating to the prevention of cruelty to animals
Republic of Africa (Republic of South Africa 1998)	1998	Animal Improvement Act, 1998	To provide for the breeding, identification, and utilisation of genetically superior animals to improve the production and performance of animals in the interest of the Republic; and to provide for matters connected therewith
Government Gazette (Center for Environmental Rights 2020)	2000	Meat Safety Act 2000	To provide for measures to promote meat safety and the safety of the animal products; to establish and maintain essential national standards in respect of abattoirs; to regulate the importation and exportation of meat; to establish meat safety schemes; and to provide for matters connected therewith
Parliament of the Republic of South Africa (Republic of South Africa 2004)	2004	National Environmental Management: Biodiversity Act, 2004	To provide for the management and conservation of South Africa's biodiversity within the framework of the National Environmental Management Act, 1998; the protection of species and ecosystems that warrant national protection; the sustainable use of indigenous biological resources; the fair and equitable sharing of benefits arising from bioprospecting involving indigenous biological resources; the establishment and functions of a South African National Biodiversity Institute; and for matters connected therewith

3 Considerations to follow to achieve an ethically conscious AI system

Integration of AI Ethics in the planning, design, and development processes of all Wildlife Conservation tools and technologies is vital. It is essential, because it minimises the level of criticism experienced by AI monitoring tools in wildlife conservation. It also fosters and creates trust between humans and machines. Due to the expansion of AI tools and technologies, AI engineers should collectively understand ethical AI principles and ideas on the practical implementation of AI Ethics into AI for wildlife conservation (Ienca and Vayena 2019).

In this study, we propose an ethical framework with guidelines that could help to reduce the failure of AI systems. The framework uses agile methodology to implement AI Ethics. The solution specifically focuses on minimising different types of bias. The developed framework establishes and provides an ethical foundation for building and using ethically aligned AI systems. The proposed approach should follow key phases in the software development life cycle. The stages must use an agile methodology that embraces the idea of iterative development while implementing the AI Ethics guidelines into AI systems (Dingsøyr et al. 2012). Agile methodology is not entirely new to the software development community and can quickly be adopted and implemented. For the study, the two agile methodologies that are being recommended are Kanban and Scrum; both are

known to lead to the development of successful projects (Lei et al. 2017). These two approaches work incrementally by optimising the development process through identifying the tasks, managing time more effectively, and setting up teams.

The framework has guidelines to follow under each entity, such as legal, data and algorithms. Figure 3 provides a comprehensive view of the entities and activities involved. In implementing this framework, stakeholders must pay attention to engaging in activities, such as:

- Conducting algorithm audits by AI developers in wildlife conservation. These monitoring tools are essential. Audits performed on algorithms must state parameters audited; one can also audit optimisation procedures, learning resources used, data sets, and mathematical models to simplify workflows in minimising bias.
- Drafting laws and policies. These laws must embrace normative values, apply transparency and prevent the cruelty of large animals. When animal lives are threatened, animal rights are violated (Gutema 2015). The laws must support the disclosure of information such as conservation laws and policies, data sets, and algorithms, which lead to perceived legitimacy (de Fine Licht and de Fine Licht 2020), since it minimises harm and improves AI (Ienca and Vayena 2019). In implementing such laws and policies, trust in AI in wildlife conservation increases, and the non-violation of animal rights, data privacy and security are attained.

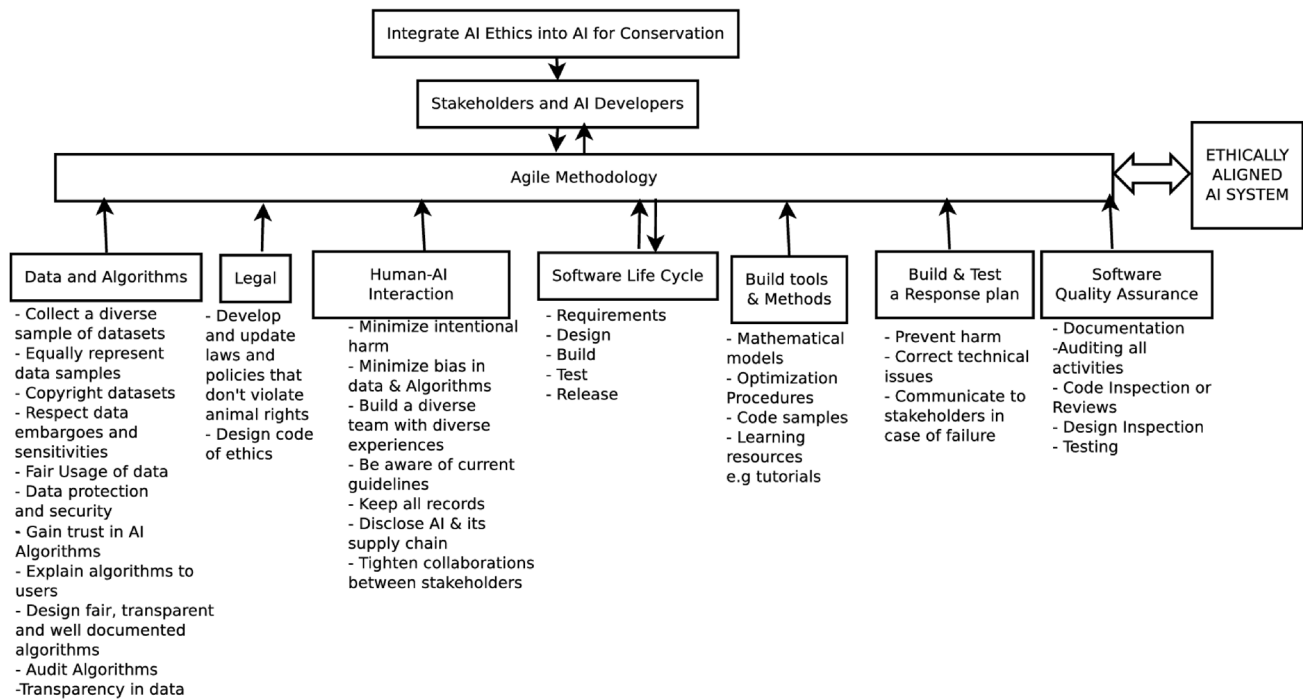


Fig. 3 Framework that guides stakeholders to implement AI ethics in AI systems in wildlife conservation

- Agile teams should design a code of ethics, software quality assurance and tighten existing collaborations between conservation policymakers and software companies. WWF is currently collaborating with Intel to apply AI to monitor and protect Siberian tigers in north-eastern China (Xu 2019).
- AI developers must explain how machine learning algorithms implemented to users work, to gain trust in these AI models. The system users must be aware that they are interacting with an AI system that is keeping track of user location data, among other data elements.
- Conclusions and recommendations of the developed and deployed AI system have to be understood and justification of a particular outcome stated (Doshi-Velez et al. 2017).
- The development team can develop safe, reliable, and robust systems that prevent failure, identify and monitor failure, and maintain the model (Saria and Subbaswamy 2019). This framework also uses an incident response plan; this plan is usually developed, adopted, and tested to prevent, identify, and monitor failures. The incidence response plan must prevent harm, correct technical issues, and communicate what happened, as well as approaches adopted to solve the problem, to customers and employees. The implementation of the incidence response plan module must follow a cycle of development, validation, testing, and refinement, to

extensively minimise the consequences. This approach has been successful in cybersecurity (Charles and Syed 2015).

The framework in Fig. 3 with its guidelines provide AI developers with an initial approach for implementing AI Ethics in current AI-based conservation systems. AI developers and stakeholders interact with the framework iteratively through activities such as data and algorithms, legal, human–AI interaction, building tools and methods, building and testing a response plan, and conducting software quality assurance. All these can be implemented using Scrum or Kanban.

When designing a completely new AI system in wildlife conservation, AI developers must use the proposed framework (Fig. 3) to create robust, safe, fair, transparent, and well-documented algorithms. Some applications on-demand in wildlife conservation, where our proposed framework can be applied, are:

- Detecting poachers and their biometrics. In catching poachers, the poachers' images are easily obtained from camera traps and used for object detection. Detection of biometrics is possible when poachers' fingerprints are collected from reliable on-site sources.
- Design AI algorithms to detect the transportation of illegally traded goods from wildlife as well as final prod-

ucts from such goods. The installation of AI platforms that can detect these illegally traded goods across border points could help in curbing this damaging trade. The algorithm should alert all wildlife conservation authorities and airport or border control authorities when the AI algorithm detects a product or by-product. If a large number of authorities get alerts about illegal goods simultaneously, this could minimise corruption and bad practices at border points.

- AI-based systems that can detect illegally traded goods from wildlife on social media platforms using our proposed framework are possible. For example, demands have been increasing from various researchers requesting Facebook to loosen its rules, to allow identification and access to Facebook groups involved in the illegal trade of wildlife products. Ongoing work on illegal trade in China via social media has shown promising results (Xu et al. 2020).

4 Conclusions

Currently, the two biggest threats to wild animals are illegal trade and poaching (Anderson and Jooste 2014; Kerry 2012). Due to these threats, a growing number of monitoring tools are being implemented in wildlife conservation, some with encouraging results. Most wildlife conservation organisations implement these solutions in harmony with existing national laws, policies, and guidelines. The latter are often drafted through collaborations. When complying with existing regulations, organisations typically attempt to minimise biases related to the unfair representation of some species. Minimising these biases is a form of compliance with ethics and AI Ethics in wildlife conservation. AI Ethics, in particular, is in increased demand in sectors ranging from health, education and e-commerce to wildlife conservation. These sectors equally aspire to implement ethically conscious AI systems that do not cause harm to stakeholders and can perform tasks while observing normative values. In the specific case of wildlife conservation, guidelines on the ethical use of AI-based monitoring tools such as drones are required. In South Africa, there is an ongoing ban of drones from flying around game reserves. These guidelines ensure that animals are protected, and their rights are not violated. Notwithstanding these stringent requirements, no existing policy and framework explicitly elaborate on the practical implementation of AI Ethics into AI-based monitoring tools in wildlife conservation. This persistence constitutes a major gap in the ecosystem of AI-driven platforms for wildlife conservation.

In this work, we performed a non-exhaustive review of 59 articles focused on exploring dominant AI Ethics and AI

Ethics issues in wildlife conservation, as well as key obligations for stakeholders in AI for wildlife conservation. In providing recommendations to mitigate the identified gaps, we designed a framework, summarised in Fig. 3, aimed at facilitating the practical implementation of AI Ethics in AI systems for wildlife conservation. This included the listing of on-demand AI systems in the sector. The documented framework and guidelines specifically aim at:

- Improving the development and deployment of machine learning models in reducing bias;
- Promoting fair and ethical use of algorithms, data, and systems;
- Explaining any decision-making process with due clarity; and
- Implementing safeguards that protect all the stakeholders of the designed system.

Among others, the proposed solution helps in minimising AI Ethical concerns in data, algorithms, legal aspects, and human–AI interaction by relying on the agile methodology; the latter being a well-tested methodology in software development. The framework is iterative and implemented incrementally using Scrum or Kanban. While developing this framework, we considered four primary goals:

- To create awareness of the guidelines of AI Ethics in wildlife conservation;
- To make the framework suitable for agile development;
- To make the proposed solution practical by having a software quality assurance and response plan that observes data protection and security, in case of failure; and
- To consider tools, methods, and techniques in the development process.

With the tested agile methodology at its core, the proposed framework has the potential to mature into established mitigation for the identified gaps. We note, however, that our study remains limited as it focused on both guidelines and practical methods for South Africa. Future work should focus on extending this review to the whole of Africa and subsequently to other continents, such as Latin America and Asia. With this in mind, the focus should also be on the practical implementation of the proposed framework or via active prototyping. We also note that the framework can also be implemented in several AI systems in sectors ranging from agriculture, health, education and manufacturing.

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